

LogitTrail: An Inspectable Local Interface for Typed Decisions from Frozen Language Models

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Abstract

Many language-model applications need a small typed decision rather than free-form text. LogitTrail maps a state and criteria to a categorical choice, a true-candidate probability, or an expected ordinal stage using frozen next-token logits, with no trained decision head. Its local browser exposes typed values and distributions; a separate model-free explorer aligns saved conditions by candidate meaning and traces results to source records and physical calls. A 4,050-request matched comparison finds identical direct-readout and native one-token outputs in all 1,350 pairs, without an established latency advantage. Two subsequent studies across three checkpoints use 7,600 and 14,400 requests to examine presentation sensitivity and probability averaging. Reduced sensitivity can coexist with incorrect or nearly constant predictions, while averaging costs multiple model calls. The contribution is a runnable inspection workflow and auditable system evaluation separating type validity, candidate concentration, task quality, and computation cost.

1 Introduction

Applications may need a route, an urgency score, or a rubric stage. Typed interfaces make these contracts explicit, but valid types and concentrated candidate probabilities do not establish correct decisions.

LogitTrail (formerly Mini Jev) demonstrates this distinction through a local browser interface, an HTTP/Python API, and a saved-evidence explorer. Users can run typed questions locally or inspect recorded comparisons without downloading a model. The interface is inspired by TypeSafe’s Jev (Almeida, 2026; TypeSafe, 2026b); this independent implementation does not reproduce its architecture, training, calibration, compatibility, or performance.

Developers need to determine what changed in a decision, whether the change improved it, and which executions support that conclusion. Our workflow connects semantic candidate comparisons to reference outcomes, panel frequencies, and source provenance. Its mechanism—frozen candidate scoring, restricted softmax, and typed response construction—is conventional (Schick and Schütze, 2021; Ma et al., 2025; LMQL contributors, 2026). Neither scoring nor typed probability interfaces are new.

Source and artifacts: github.com/UpHash-Network/mini-jev. Browser access: [saved-evidence explorer](#). Video: [captioned demonstration](#).

2 System and Typed Contract

2.1 From a state to typed values

A request contains a shared state and named questions. Each question supplies a type, natural-language instructions, and candidate criteria. The service returns a semantic label, a candidate distribution, and a type-specific value:

- **Choice** returns the key of the most likely candidate. Keys are canonically sorted before prompting.
- **Noul** exposes the probability of the true candidate in a two-candidate false/true decision.
- **Score** returns the expected zero-based stage of an ordered rubric; the most likely stage is also available as the label.

The distinction between Score’s expectation and its hard label matters. An expectation of 1.6 on stages 0–3 is a probability-weighted position, not an additional model-generated stage or a calibrated utility. Its interpretation assumes equally spaced stage indices; users must decide whether that fits their rubric.

Let z_i denote the next-token logit for candidate $i \in \{0, \dots, K-1\}$, with K candidates and temperature $T > 0$. The service computes

$$p_i = \frac{\exp(z_i/T)}{\sum_{j=0}^{K-1} \exp(z_j/T)},$$

$$s = \sum_{i=0}^{K-1} ip_i. \quad (1)$$

These are probabilities conditional on the selected candidate set. They are neither probabilities over all valid textual answers nor guarantees that the criteria are exhaustive. Changing the criteria changes the question as well as the normalization. The optional scalar temperature preserves the maximum-logit label, but can change Noul and Score values.

The interface also displays entropy concentration,

$$c = 1 - H(p)/\log K,$$

$$H(p) = - \sum_i p_i \log p_i. \quad (2)$$

A concentrated distribution has c near one. The display identifies this quantity as concentration rather than estimated accuracy. The response includes probability-semantics and calibration metadata, so downstream applications can retain the same distinction.

2.2 Local execution

The released inference configuration uses Qwen3.6-35B-A3B in Q4_K_M GGUF form (Qwen Team, 2026; ggml-org, 2026), through a source-built llama.cpp helper (ggml-org and contributors, 2026). The model and its output projection remain frozen; there is no trained decision head. The helper performs the full vocabulary projection and then gathers candidate logits. It does not implement a selected-row projection optimization.

The prompt repeats the task twice, following the use of repetition for non-reasoning inference (Leviathan et al., 2025). Choice and Noul use letter labels; Score with at most ten stages uses digits after a prefilled answer-object prefix. Larger Score rubrics use letters. The implementation checks that every candidate is a distinct single token at the actual prefix boundary. The API accepts 2–26 candidates; quality evaluation covers 2–8, while larger candidate sets have functional coverage only.

A multi-question request executes its questions sequentially. Model state is cleared between questions: they share the supplied state text, but not a cached prefill. Complete inputs exceeding 2,048 tokens are rejected rather than silently truncated. Schema validation and deterministic response construction prevent free-form output from crossing the API boundary; these software properties do not prevent semantic errors.

2.3 Relationship to one-token decoding

Restricted softmax readout has the same mathematical candidate distribution as a sampler that masks all non-candidates, provided the model prefix, logits, candidate IDs, temperature, and other processors match. Greedy labels then agree unless tie-breaking differs; stochastic sampling agrees in distribution, not necessarily in realized samples. This is an equivalence under specified conditions, not a new inference theorem or learning method.

In particular, selecting a first token after prefill need not require another model forward pass. Avoiding a generated token therefore does not itself establish a speedup. The matched study in Section 4 tests this point using an actual native one-token sampler, rather than comparing direct readout only with a long textual answer.

3 Demonstration and Access

3.1 Inspect saved evidence without a model

The static [saved-evidence explorer](#) requires only a browser. It contains all eligible records from the presentation and averaging studies: 22,000 physical calls and 9,000 derived answers in 18 panels. Their 3,600 study/item/model combinations represent 1,200 distinct source items, reused across checkpoints. These are recorded observations, not new inference.

Users choose a study, checkpoint, task, item, and two conditions. Both sides retain the same item and model; probabilities align by semantic key on a common 0–100% scale even when token bindings or row order differ. The display separates the selected label, typed value, concentration, and reference. Full-panel label frequencies remain visible independently of item selection. This permits inspection of an order flip, a correct answer lost after averaging, or a nearly constant predictor. Four outcome-selected examples illustrate these cases; they do not estimate prevalence.

The explorer also distinguishes executions from

derived results. An averaged answer lists its physical members; comparisons count their union without double-counting shared calls. Source details identify archive members, JSONL lines, mappings, and hashes. Loaded panels are SHA-256 checked; comparison exports retain exact values and sources. Original task text is withheld. URL fragments share selections. Software checks match all exported records, 172,200 dependent comparison pairs, and 155 panel distributions to raw calculations. A complete-JSON view retains the same information; these are not human-performance results.

A separate [local JSONL importer](#) aligns user-supplied physical logits or declared full probabilities and exports typed comparisons without uploads or storage. Missing references/latency remain unknown; source authenticity and derived-member recipes are unsupported. Synthetic tests and one hash-selected archived item check compatibility, not practical advantage.

3.2 Run local typed decisions

The separate live browser connects through a Python bridge to the local model API. Users edit a shared state, instructions, and criteria for Choice, Noul, and Score, then select *Run decisions*. Results expose candidate probabilities, concentration, elapsed time, and request/response JSON. Editing marks prior results stale. The live UI has no replay or cyclic-averaging mode: an unavailable API produces an error. Input edits explore behavior, not controlled accuracy.

A linked [reviewer guide](#) proceeds from the explorer to model-free verification and optional live installation. The tested live environment is a 64 GB Apple M5 Pro; its model file is approximately 20.4 GB. Appendix A gives installation and verification boundaries. The static explorer has no model, API-key, or build requirement.

4 Evidence Behind the Demonstration

4.1 Evaluation panels and provenance

Table 1 separates the evidence sources. Repeated requests are not additional quality examples, and the external tasks do not share one accuracy metric.

The local regression suite has 2,220 generated items from 45 template families and 180 individually AI-authored items; its legacy manual field does not mean human annotation. Earlier errors

Evidence	Size	Interpretation
Local regression	2,400 items	Self-authored Japanese; 800 per type
Expanded external	600 items	JCoLA, JSTS, JCQA; 200 each
Matched runtime	4,050 quests	re- 150 local $\times$ 5 repetitions, plus 600 external; 3 modes
Presentation study	7,600 quests	re- Same 600 external items; 3 checkpoints
Cyclic averaging	14,400 requests	600 additional external items; 3 checkpoints
Diagnostic replay	2,400 quests	re- 240 further JCQA items; 2 checkpoints
Scope extension	4,800 quests	re- 240 new JCQA + 240 English items; 2 families

Table 1: Evaluation accounting. The original panels use 1,200 items; two later diagnostic studies add 240 and 480. Measured counts exclude synthetic checks and warm-ups. JCQA abbreviates JCommonsenseQA.

informed authoring. All 2,400 outputs were valid, but 162 labels disagreed with authored references. Temperature fitting did not uniformly improve calibration or score error.

The expanded sample has 200 public-development examples each from JCoLA (Someya et al., 2024), JSTS, and JCommonsenseQA (Kurihara et al., 2022). Hash selection and rubrics were frozen before inference, without independent preregistration. Six JSTS rows share text or image provenance with an earlier 300-item JNLI pilot. Public development data may have appeared in model training; these are not independent hidden-test evaluations.

4.2 Matched runtime and parity

All three conditions used one resident model and a common research HTTP response containing a type and semantic label. Direct readout and native one-token greedy selection used identical complete prompts and candidates. The JSON condition actually generated a grammar-constrained object, with a serialization-specific prompt. Its timing comparison therefore includes prompt and pre-fill differences as well as multi-token generation; it does not isolate sampler overhead.

All $(150 \times 5 + 600) \times 3 = 4,050$ requests completed without recorded failures; 21 synthetic warm-ups are excluded. Mode order was randomized within item/repetition. Research-harness timing spans serialization through loopback HTTP parsing, including audit transfer but excluding trace writes and browser rendering.

Direct and one-token conditions matched

Candidate comparison

[Open complete JSON for this selection](#) - Includes the selected comparison, full panel, and source index.

A Recorded condition

Original presentation

The saved original presentation.

B Recorded condition

Reversed candidate rows

Reverse only the displayed rows, preserving answer-token, key, and meaning bindings.

The most likely answer stays the same. Largest candidate probability change: 7.384769 percentage points.

Most likely semantic answer

A true
B true

Recorded tie handling is preserved.

True-candidate probability

A 0.986484
B 0.912636

Within the false/true candidate pair.

Concentration

A 0.8967
B 0.5724

0 = uniform - 1 = peaked; not accuracy.

Model calls per result

A 1
B 1

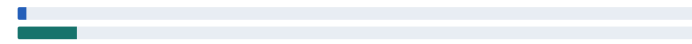
0 shared; 2 distinct calls across A and B.

Probability within the candidate set

A and B share a 0–100% scale. Δ is B – A in percentage points. Values are aligned by semantic key, even when presentation order changes.

false · unacceptable
key: false

A
B



1.351581%
8.73635%
 Δ percentage points

true · acceptable
key: true

A
B



98.648419%
91.26365%
 Δ percentage points

Dataset reference label: true. A matches; B matches. This is one recorded item, not a new accuracy estimate.
► Answer frequencies across this full panel

Concentration measures how peaked a candidate distribution is; it is not the probability of being correct. Score expectations assume equally spaced stages. Recorded examples do not establish human usability or a causal model-size effect.

Figure 1: The saved-evidence explorer aligns recorded candidate probabilities by meaning. Users inspect labels, typed values, references, and physical-call costs, then trace or export the comparison. The displayed example is an illustration, not a representative performance sample or a live model execution.

Comparator — direct	Difference	95% interval
One token	+0.0898	[−0.7483, 0.8449]
JSON	+160.2958	[138.1189, 177.0587]

Table 2: Primary paired HTTP latency differences in milliseconds. Intervals resample observed local family/tag clusters; JSON uses a different serialization prompt.

prompts, token sequences, candidate IDs, logits, and labels in all 1,350 paired observations. Maximum observed candidate-logit difference was zero; there were no maximum-logit ties. The primary latency statistic is the median across 150 local items of the within-item difference between five-repetition mean latencies. It is not a difference between marginal medians.

Table 2 uses 10,000 whole-family/tag percentile bootstraps, conditional on this model, machine, items, and session. The one-token interval crosses zero; JSON is slower under different serialization requirements. These intervals do not estimate

between-machine or independent-session variability.

4.3 External task-specific results

Direct readout and one-token selection have identical external labels. JCoLA reached 86.0% accuracy and MCC 0.5708; JSON reached 82.0% and MCC 0.5160. The selected sample contains 158 acceptable sentences, so an always-acceptable rule already achieves 79.0% accuracy and has undefined MCC. Reporting only accuracy would obscure this imbalance.

JCommonsenseQA accuracy was 95.0% for direct/one-token and 95.5% for JSON, a difference of one example. JSTS retains its continuous reference scores in $[0, 5]$: direct expected-stage MAE at $T = 1$ was 0.5065. JSON has no comparable probability vector, so we do not assign it an expected-stage metric. The separate hard-stage comparison gives MAE 0.5480 for direct/one-token and 0.5650 for JSON. We neither round gold scores into classes nor pool task metrics.

4.4 Presentation sensitivity and averaging

Two subsequent studies add Qwen2.5-0.5B-Instruct and Qwen2.5-1.5B-Instruct to the native 35B checkpoint. The first uses 7,600 requests on the same 600 external items. All 1,800 exact-input repeat pairs have identical logits and probabilities, yet display reversal changes native JCQA labels on 11/200 items. Conversely, the 1.5B checkpoint has zero JCoLA flips while always predicting acceptable: stability alone is insufficient.

The second freezes 600 additional public-development items before its 14,400 requests. It excludes prior IDs and literal text reuse, and selects JSTS pairs from previously unused sentence/image components. However, 194/200 JCoLA items share bibliographic sources with earlier evaluations. Both studies complete without inference errors; each excludes 57 synthetic checks/warm-ups.

We average semantic candidate probabilities at $T = 1$ across cyclic rotations of complete option lines, keeping output-token-meaning bindings fixed. Fixed-binding transformations and cyclic aggregation have precedent (Wei et al., 2024; Zheng et al., 2024); this is not a new algorithm or a reproduction of PriDe. Table 3 compares canonical-single and forward-cyclic accuracy, alongside single/reversed and forward/reverse-cyclic label disagreement. Lower disagreement does not consistently improve accuracy. The 0.5B cyclic JCoLA output is always acceptable (78.5% accuracy, undefined MCC). On JSTS, 0.5B and 1.5B cyclic hard labels occupy one stage on 199/200 and 200/200 items, respectively. These marginals expose collapse hidden by type validity or stability.

One cyclic answer costs 2, 5, or 6 model calls for JCoLA, JCQA, or JSTS; the single baseline reuses a pool member. Binary forward/reverse pools contain the same two calls, so their equality is structural. There is no equal-call repeat control in this averaging study. The small checkpoints use Transformers/MPS float32; the native model uses quantized Metal inference. Architecture, precision, and backend differ, precluding a model-size or cross-backend speed conclusion. These are descriptive system checks; fuller distributions, grouped intervals, and call accounting accompany the artifacts.

Diagnostic value. A frozen 240-question JCQA panel adds 2,400 calls across Qwen 1.5B and

Checkpoint	Accuracy (%)	Flip rate (%)
Qwen2.5 0.5B	39.5 → 36.0	65.5 → 30.5
Qwen2.5 1.5B	80.0 → 83.5	26.0 → 12.5
Qwen3.6 35B-A3B	98.5 → 98.0	4.5 → 1.5

Table 3: JCQA on the additional 200-item panel: single → cyclic point estimates. Flip rate measures orientation disagreement, not error rate. Native 35B uses Q4_K_M. These are not equal-compute comparisons.

35B. Adding order-induced total variation (TV) to entropy did not establish added error-detection value; at 720 calls, the two policies have identical per-item correctness. A separate scope extension adds 4,800 calls on 240 new JCQA and 240 English CommonsenseQA items (Talmor et al., 2019), shared by Qwen 1.5B and Phi-4-mini (Microsoft, 2025). The extension also does not establish added order-TV value: the primary Phi-English AP difference is -0.051 ; at 720 calls, both policies return 174/240 correct answers. Full intervals and all budgets are retained (Appendix A, Table 5).

4.5 Reproduction and framework integration

The versioned source/evidence package at github.com/UpHash-Network/mini-jev records the three studies’ 26,050 requests, protocols, selections, hashes, analyzers, and AI-agent audits. Clean-directory CPU reanalysis reproduced all 56 derived files byte-for-byte. A fresh native build served all three types using a cached, hash-verified model on the same Mac. This validates reconstruction with source/model cache reuse, not a second-machine or fresh-download replication.

Separate synthetic integration checks retain failures. With matched checkpoint, prefixes, and candidate IDs, LMQL 0.7.3 through a custom native backend matched all 12 labels, but three probability differences exceeded 10^{-5} (maximum 2.63×10^{-5}). ChainForge 0.3.7.4’s custom provider preserved eight typed-answer pairs and rejected three invalid inputs. These checks concern interoperability, not human usability or task quality; Appendix A supplies the scope and post-hoc analysis. Independent cross-machine inference remains untested.

5 Related Work and Scope

Candidate and ordinal scoring. Verbalizers (Schick and Schütze, 2021), decoding-free selection (Ma et al., 2025), Qwen3 reranking (Zhang

et al., 2025; Qwen Team, 2025), and G-Eval’s weighted scores (Liu et al., 2023) precede our readouts. Zhuang et al. (2024) compute expected relevance from normalized fine-grained label log-likelihoods in zero-shot ranking. Wang et al. (2026) use grade-token expectations for job-candidate ranking and fine-tune with a combined mean-squared-error and cross-entropy loss. Score shares this established expectation; frozen-weight scoring is also not new. Our contribution concerns typed inspection and system evaluation, not ranking gains or a new scoring rule.

Structured interfaces. LMQL exposes continuation scoring and selection (LMQL contributors, 2026); multi-token scoring need not equal our single-token readout. Outlines, Guidance, and grammar-constrained decoding support structured generation (dottxt and Outlines contributors, 2026; Guidance contributors, 2026; Geng et al., 2023). TypeSafe’s adapter maps general LLM outputs to a Jev-style contract (TypeSafe, 2026a); generated probability values differ from token-derived candidate probabilities. Our JSON experiment is a serialization comparison, not a framework benchmark.

Inspection interfaces. Rakutko’s distinct minijev compares frozen letter-logit scoring with constrained JSON in an inspection bench (Rakutko, 2026); our rename avoids confusion. Ollama documents local Choice/Noul/Score probabilities (Ollama, 2026); ChainForge adds an Ollama decision provider (Arawjo et al., 2024; ChainForge contributors, 2026). Its inference path remains untested here. ChainForge 0.3.7.6’s GUI retains six matched saved-evidence cases; its native Processor also executes our custom JavaScript to reconstruct 2,400 conditions from 6,000 archived logit records within 10^{-12} tolerance. This supports competitor feasibility, not inspection superiority (Appendix A). LLM Comparator supports response comparisons (Kahng et al., 2025); Langfuse exposes typed distributions (Langfuse, 2026), and Phoenix preserves Choice metadata (Arize AI and Phoenix contributors, 2026). Our distinction concerns connecting complete panels, semantic alignment, shared calls, and archive-line provenance in one workflow, not typed inspection alone.

Sensitivity and calibration. Order/token bias and fixed-binding changes are established (Wei et al., 2024; Zheng et al., 2024). Preprints sep-

arate order stability from accuracy (Hanna and Feng, 2026), ordinal hard-label collapse from concentration (Gao et al., 2026), and correlated judge errors from confidence-cascade savings (Rao and Callison-Burch, 2026). These overlap our observations; our studies are not reproductions or claims of new general phenomena. Temperature scaling (Guo et al., 2017) differs from contextual label-bias correction (Zhao et al., 2021); canonical sorting does not establish semantic or position invariance.

Limitations

Presentation studies cover three checkpoints on one Apple Silicon device; matched parity/latency covers one native quantized model in one session. Architecture, precision, and backend differences prevent model-size attribution. Public subsets have unknown training contamination and source dependence. The English extension covers one five-choice task; model sizes and languages are confounded with family and dataset. Local data are AI-authored and development-related, without independent human reference validation. No user study establishes usability or downstream decision quality.

Candidate probabilities vary with prompts, labels, coverage, and temperature. Concentrated wrong answers occur; type validity and entropy concentration cannot replace application-specific validation before consequential actions. Score assumes equally spaced stages; Noul is a relative candidate probability, not a verified truth frequency. Questions run sequentially on a large model; other operating systems and lower-memory configurations remain unvalidated. Live-demo timings are not directly comparable with the research harness’s controlled interval.

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Ethics and Availability

The demonstration supports model inspection and typed-interface prototyping, with no evidence for high-stakes deployment. Inference is local; users remain responsible for inputs and downstream use. Code and self-authored local data use MIT terms. External-dataset-derived selection metadata, rubrics, protocols, and results separately retain applicable CC BY-SA 4.0 attribution and share-alike terms. Upstream dataset text, model weights, and compiled binaries are not bundled; repository data cards and notices specify acquisition and attribution.

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A Comparison Scope and Evidence Access

Comparison	Shared conditions	Recorded observation	Unmeasured or different
Native one-token	Checkpoint, full prefix, candidate IDs/logits, greedy selection	1,350/1,350 labels match; paired latency interval includes zero	Other machines and independent sessions
Grammar JSON	Resident model, selected tasks, HTTP timing interval	Validity, task-specific quality and serialization latency	Prompt and generated length differ; no probability-vector parity
LMQL 0.7.3	Checkpoint, full prefix, candidate IDs; custom native backend	12/12 labels match; 3/12 probability comparisons exceed 10^{-5}	Native evaluation paths differ; not a stock-backend benchmark
ChainForge 0.3.7.4	Same local API, request content and typed-response contract	Eight answer pairs identical; three invalid inputs rejected	Browser flow editor and human performance
ChainForge 0.3.7.6 GUI	Six saved cases; separate raw-logit/recipe input with custom JavaScript	Saved exports agree; native Processor reconstructs 2,400 conditions within 10^{-12}	Custom diagnostic code; no inference or human performance
Langfuse Phoenix	/ Primary-source feature descriptions only	Typed evaluation and probability inspection/metadata precedents	No matched inference, GUI trial or comparative user outcome

Table 4: Evidence tiers, not a capability ranking. Unmeasured properties are not absent capabilities. Framework diagnostics are separate from the task-quality panels.

Public and internal inspection. The released HTTP response and browser expose semantic labels, typed values, candidate probabilities, concentration, and probability-semantics metadata. Raw candidate token IDs and logits belong to the internal engine and research traces; they are not fields in the public response. Table 4 distinguishes controlled comparisons, integration checks, and source-based related work. These observations do not establish that people work faster or make better decisions with this interface.

Model-free evidence review. The frozen source/evidence ZIP retains the original 26,050 measured requests; 300 earlier pilot predictions are background and excluded. Standard-library checks verify accounting, and CPU replay reproduces 56 derived files without models or network access. The current manuscript and later studies are separate packages. The explorer validates unchanged source hashes, semantic mappings and physical-call unions. Reanalysis verifies calculations, not observation authenticity or generalization.

Live installation. The repository supplies native build and pinned-model acquisition commands: `build_native.sh`, `fetch_native_model.py`, and `./run.sh -without-calibration`. Running `python3 -m demo.server -port 8766` serves the interface at <http://127.0.0.1:8766/>. Fresh builds use $T = 1$; historical calibration artifacts are bound to their recorded runtime. The tested

Mac uses macOS 26.4, Metal, and Accelerate. The 20.4 GB model needs additional runtime memory. We distribute source, not weights or binaries; a minimum-memory configuration is unestablished. The static explorer can instead be served from docs/ with Python’s HTTP server; HTTPS or localhost enables browser hash verification.

Post-hoc LMQL diagnosis. All three original 10^{-5} probability failures remain. Maximum Score expectation error is 7.18×10^{-5} , within the separate 10^{-4} typed-value tolerance. Native probability margins exceed 0.9799, so matching labels provide no near-boundary evidence. Softmax of saved LMTP logits reproduces LMQL probabilities within 3.64×10^{-9} ; native-path logit differences can account for discrepancies, but their cause remains unisolated. Type, candidate count, and prefix are confounded. This analysis adds no inference and changes no threshold.

Matched saved-evidence workflow. The [saved-evidence package](#) freezes six ID-hash-selected cases and 32 questions before GUI execution; earlier outcomes were known. All presenters receive complete panels and source indices. ChainForge’s native search, lightboxes, spreadsheet and flow exports retain evidence supplied by the adapter, which computes alignment, deltas and call unions. No human performance was measured.

Reconstruction from raw records. The [raw-record package](#) instead supplies logits, mappings, temperatures, gold references and member recipes. ChainForge executes custom JavaScript from an empty cache in Firefox 153, reconstructing 2,400 conditions from 6,000 records. An independent Python oracle agrees within 1.78×10^{-15} . Six tuple-reversal controls preserve results; eight malformed inputs are rejected. These software checks establish competitor feasibility with custom code, not human advantage or new inference. The later physical-record importer does not ingest this fixture’s derived-member recipes.

Error detection and budgeted replay. The [original diagnostic package](#) separates inspected exploration from 240 new JCQA questions and 2,400 calls. The [scope extension](#) adds distinct Japanese/English 240-item panels and 4,800 calls across two families. Selection, prompts, models and analyses were locally frozen before benchmark inference; prior null results were known. Neither study is independent preregistration or a contamination-free test. The extension uses float32/MPS, stock full-position vocabulary projection and candidate gather for both models. Its initial Phi optimization gate failed before any benchmark call. A versioned amendment retains that failure and uses full-position repeatability checks; no tolerance was relaxed. There are 128 excluded synthetic forwards across its initial failure, diagnosis and completed run.

Error detection targets an incorrect two-call mean answer. The combined score averages within-stratum midranks of normalized mean-distribution entropy and TV, $\frac{1}{2} \sum_i |p_i - q_i|$. Its entropy-only reference also costs two calls. For batch allocation, both pay two calls per question, then complete five-call means for their top 80 items: $2(240) + 3(80) = 720$. Gold and future members never enter selection. The extension’s primary stratum is Phi-English; other strata are secondary. Tasks and unequal model sizes preclude causal language/family attribution.

Both packages retain two error targets, eight scores and four policies at five budgets. The 2,000 paired-question percentile bootstraps condition on observed ranks/allocations, with unadjusted descriptive intervals and undefined counts retained. Collection pays all five calls; replay costs sum the chosen physical members. Recorded token/latency sums are not online tim-

Checkpoint/task	Errors	AP difference [95% interval]
Q1.5 / old JA	52	−0.024 [−0.092, 0.067]
Q35 / old JA	5	−0.102 [−0.464, 0.124]
Q1.5 / new JA	40	−0.048 [−0.151, 0.039]
Phi3.8 / new JA	37	−0.067 [−0.141, 0.014]
Q1.5 / EN	74	−0.032 [−0.082, 0.022]
Phi3.8 / EN	62	−0.051 [−0.119, 0.013]

Table 5: Entropy-plus-TV minus entropy AP; 240 questions per row. Old JA is the earlier panel. New JA and EN are separately shared across the extension models, not paired translations. Q denotes Qwen; Phi is 3.8B.

ing or speedups. Reanalysis is not independent inference or human validation.

Human evaluation boundary. No human study was conducted. An earlier synthetic-page rehearsal is material QA, excluded from experimental results. A counterbalanced pilot packet targets the released local importer and complete comparison JSON, with no participants or observations. Human comparative performance and cross-machine replication remain open.